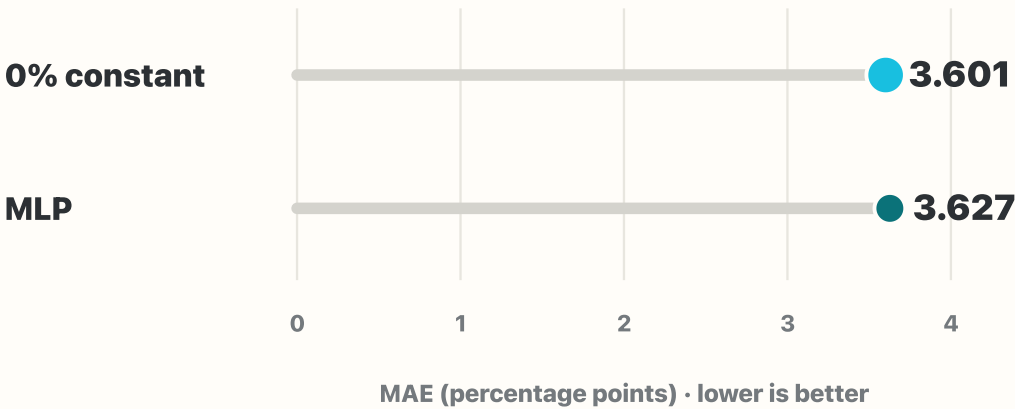
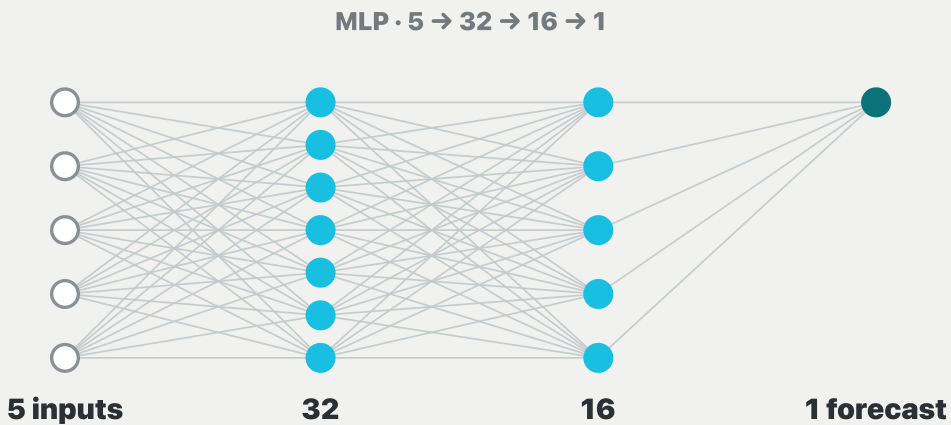


# I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.  
"Predict 0% growth" still beat it on mean absolute error (MAE).

EXPLANATION OF THE TARGET VARIABLE  
NEXT-QUARTER DOMESTIC DEPOSIT GROWTH

|                |                     |
|----------------|---------------------|
| Deposits       | \$100m → \$96m      |
| Actual growth  | -4%                 |
| Zero forecast  | 0%                  |
| Absolute error | 4 percentage points |



THE CONSTANT PREDICTED 0% GROWTH FOR EVERY BANK.

# THE BANK CAN LEND OUT MONEY YOU DEPOSIT. IT DOESN'T KEEP IT ALL IN CASH.

WHEN YOU WITHDRAW, SOME FUNDS MAY STILL BE TIED UP IN LOANS.



CASH IN THE VAULT

**\$10**



MONEY ALREADY OUT AS LOANS

**\$90**

A customer withdraws \$15. Cash is \$5 short.

**THE BANK'S TREASURY TEAM MUST FIND THE CASH.**

## HOW MUCH CASH IS ENOUGH?



Required reserve ratio: 0%. That does not mean unlimited lending.

There is no universal “keep 10%, lend 90%” rule. Capital, liquidity and other limits constrain lending.

The full liquidity coverage rule requires eligible liquid assets covering **100% of 30-day stressed net cash outflows.**

CASH

DEPOSITS

LOANS

CAPITAL

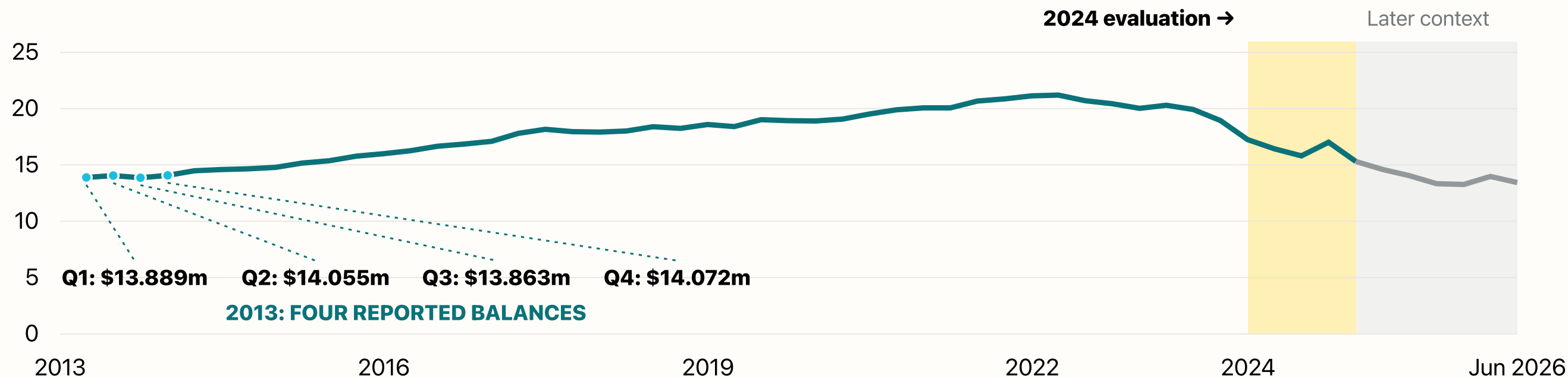
**CUSTOMERS DEPOSIT AND WITHDRAW MONEY.  
I'M FORECASTING THE NET CHANGE NEXT QUARTER.**

Example only. Reserve requirement ratios are 0%; separate liquidity rules vary by bank. Liquid assets include eligible securities, not just cash. Source: Federal Reserve, Regulation WW. This project forecasts deposit growth, not bank failure.

# "PREDICT 0%" WON. IS THIS THE GRAPH YOU WERE EXPECTING?

2013–2026 · ONE-BANK ILLUSTRATION. Zero won MAE across all test banks, not necessarily this bank.

YOUNG AMERICANS BANK · DOMESTIC DEPOSITS (\$ MILLIONS)

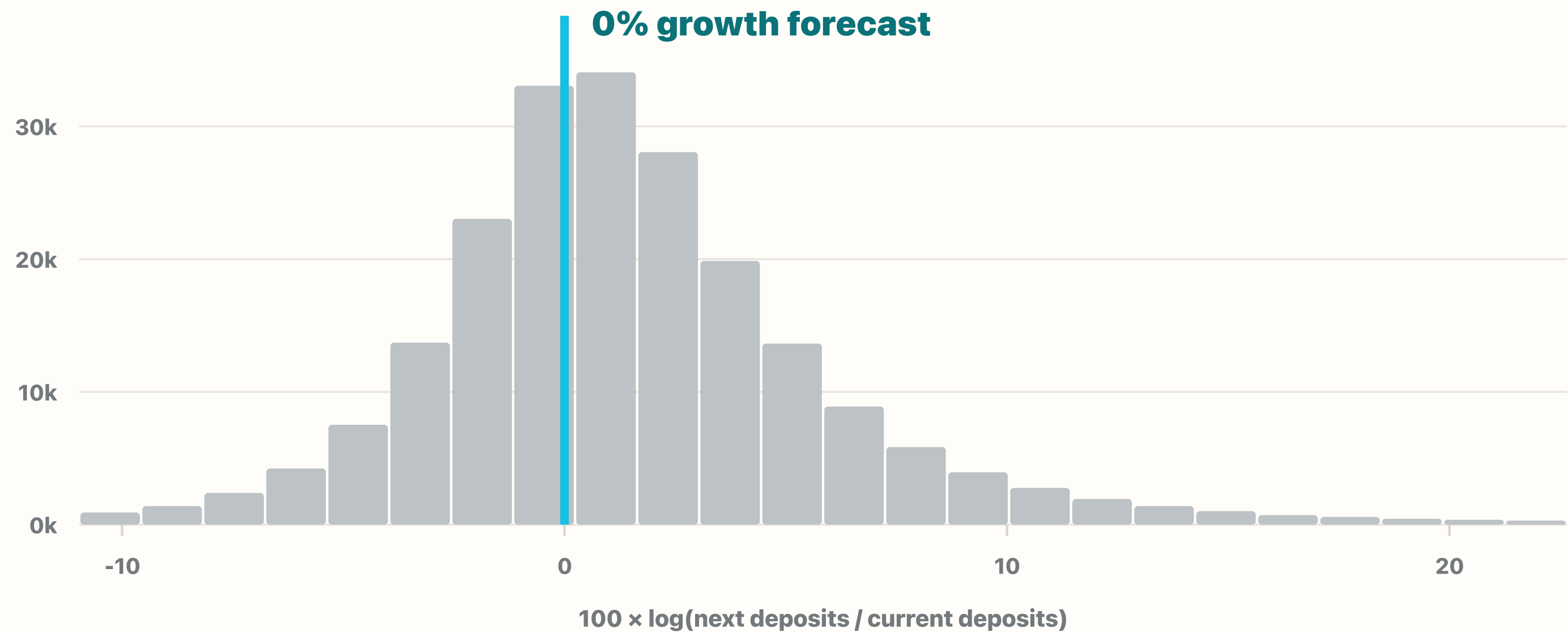


**0% growth means "same balance next quarter."**  
Look at 2013: about \$14 million, quarter after quarter.  
"Same again" would be a close guess.

**Was little change common across the other banks too? →**

Source: FDIC BankFind, Young Americans Bank (CERT 27010), quarterly domestic deposits, 2013–Jun 2026.  
2013 is a deliberately stable example. Other years changed more. Dotted lines connect labels to reported balances.

# SO WHY WAS 0% GROWTH SO HARD TO BEAT?



**A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.**

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

# BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

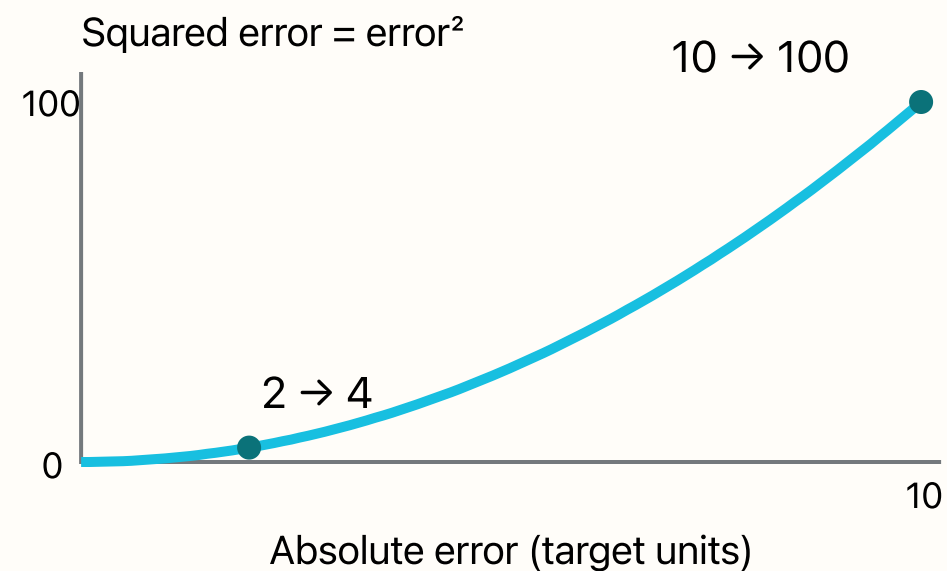
An error 5× as large gets 25× the squared-error penalty.

## FIXED BENCHMARK

### 0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.  
It never adjusts to reduce an error.



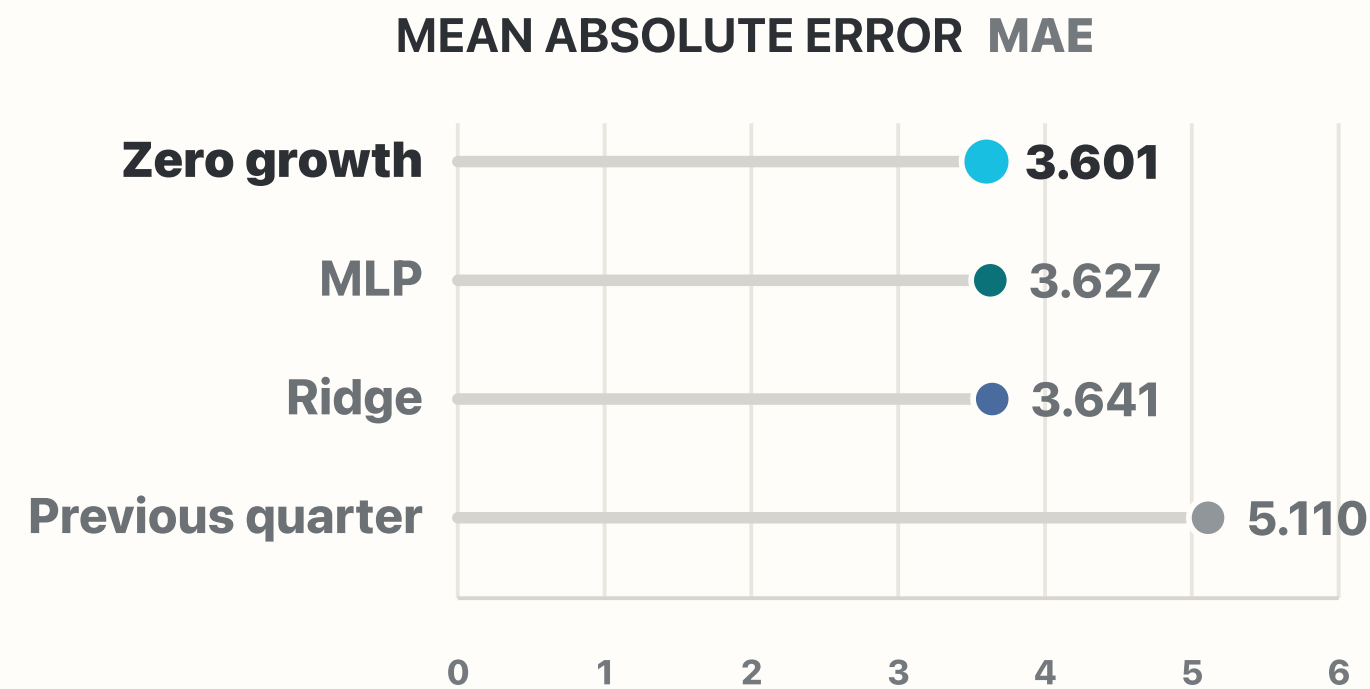
## LEARNED MODELS

### RIDGE + NEURAL NET

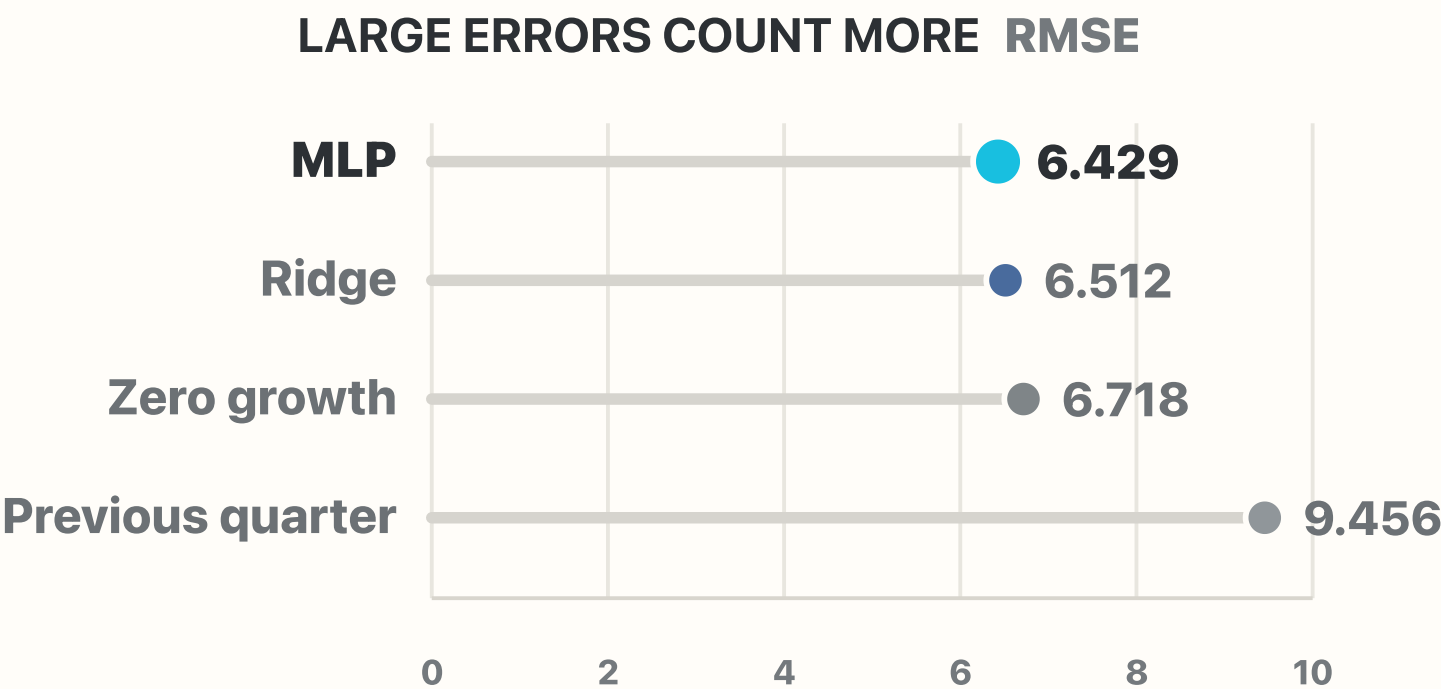
Can move away from 0% to reduce those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH  
A SLIGHTLY HIGHER MEAN ABSOLUTE ERROR.**

# ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



MAE (percentage points) · lower is better



RMSE (percentage points) · lower is better

**“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.**  
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

# 0% TIES EVERY BANK.

## THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

★ I am not a financial advisor. I do have a sorted table.  
In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%.  
Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

| RANK | BANK                       | PREDICTED GROWTH | 0% BASELINE |
|------|----------------------------|------------------|-------------|
| 01   | YOUNG AMERICANS BANK       | -4.02%           | 0% · tied   |
| 02   | PIONEER COMMERCIAL BANK    | -3.32%           | 0% · tied   |
| 03   | SUMITOMO MITSUI TR BK U SA | -2.83%           | 0% · tied   |
| 04   | OLD GLORY BANK             | -2.75%           | 0% · tied   |
| 05   | TURTLE MOUNTAIN STATE BANK | -2.11%           | 0% · tied   |
| 06   | COLUMBUS STATE BANK        | -2.05%           | 0% · tied   |
| 07   | FDS BANK                   | -2.03%           | 0% · tied   |
| 08   | BOELUS STATE BANK          | -1.96%           | 0% · tied   |
| 09   | LEMONT NATIONAL BANK       | -1.80%           | 0% · tied   |
| 10   | CITIZENS NB OF CROSBYTON   | -1.75%           | 0% · tied   |

GOAL: FIND MORE LOWEST-GROWTH BANKS WITH THE SAME 454 REVIEWS.

HIGHER IS BETTER.

Random selection

45.4 expected

Linear model (Ridge)

112 found

Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth.  
112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

# I THREW OUT 4 INPUTS.

## KEPT DEPOSIT SIZE ALONE.

Do cash, loans, equity, and prior growth help choose which banks to review?

### REVIEW-LIST TASK: FIND MORE LOW-GROWTH BANKS · HIGHER IS BETTER



Banks in next-quarter lowest-growth 10% per 100 selected

### FORECAST-ERROR TASK: MAE · LOWER IS BETTER

| Predict 0% | Neural network | Five-input Ridge | Deposit-size Ridge |
|------------|----------------|------------------|--------------------|
| 3.601 pp   | 3.627 pp       | 3.641 pp         | 3.654 pp           |

Zero has the lowest MAE here. It gives every bank the same forecast, so it cannot order a review list.

Ranking: equal-quarter averages, three reused 2024 quarters. Size-only challenger added after examining outcomes. Original evaluation population.

# I KNOW WHAT YOU'RE THINKING.

## WHY NOT JUST USE ANOMALY DETECTION?

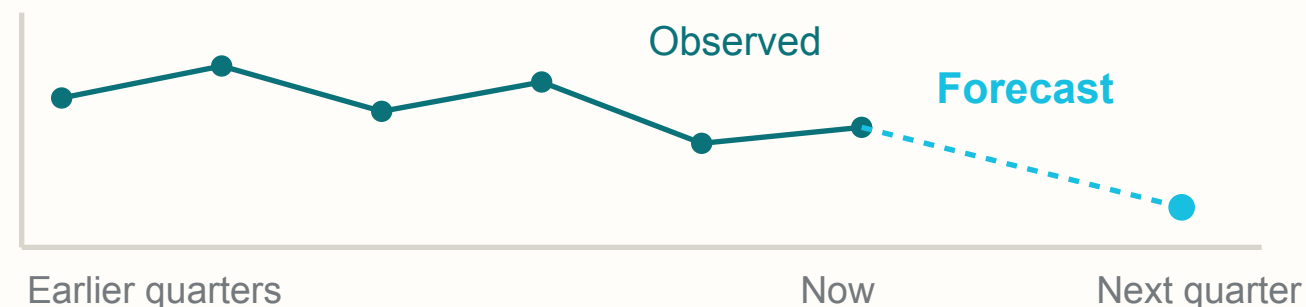
Find the unusual banks. Put them on the review list. Isn't that what we're looking for?

**Well, let's try it.**

- Eight quarters of the same five financial inputs.
- Three methods: PCA, Isolation Forest, and a small autoencoder.
- Same banks to choose from. Same 10% review capacity.

### FORECAST: predict the next value

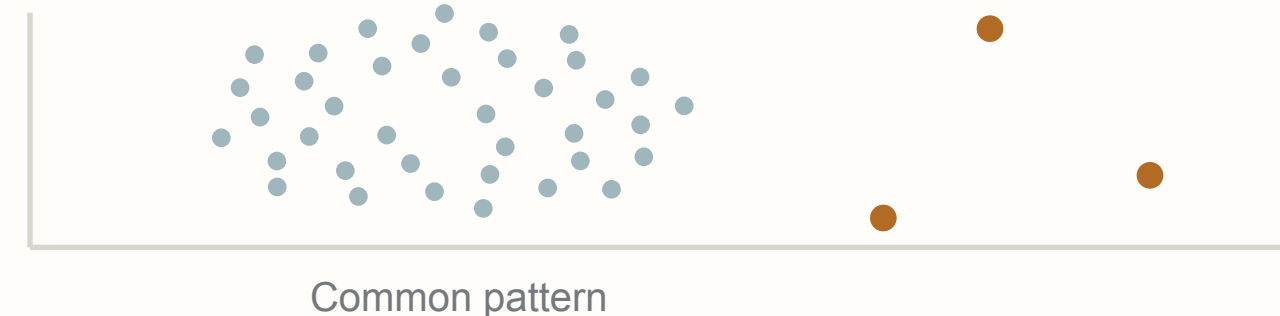
Schematic deposit history



**Select the 10% with lowest predicted growth.**

### ANOMALY: find unusual observations

Schematic feature-space view



**Select the 10% with highest anomaly scores.**

Same workload: choose 10 of every 100 banks. Then check their next-quarter growth.

**Would choosing unusual histories find more low-growth banks?**

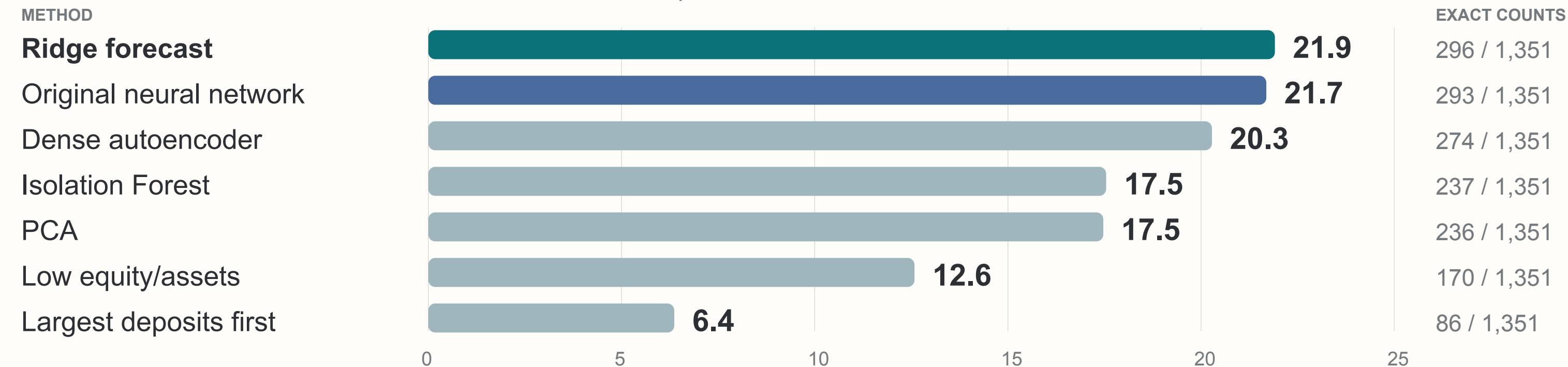
Illustrative mini-plots, not measured model outputs or a claim that the detectors use clustering. Forecast inputs include prior growth.

# RIDGE STILL BUILT THE BEST REVIEW LIST. MORE LOW-GROWTH BANKS. SAME WORKLOAD.

STILL THE REVIEW-LIST TASK: cases found per 100 selected. HIGHER IS BETTER.

Out of every 100 selected, how many landed in the lowest-growth 10% next quarter?

All seven methods: 1,351 selections each across three 2024 quarters.



## SCORECARD SO FAR

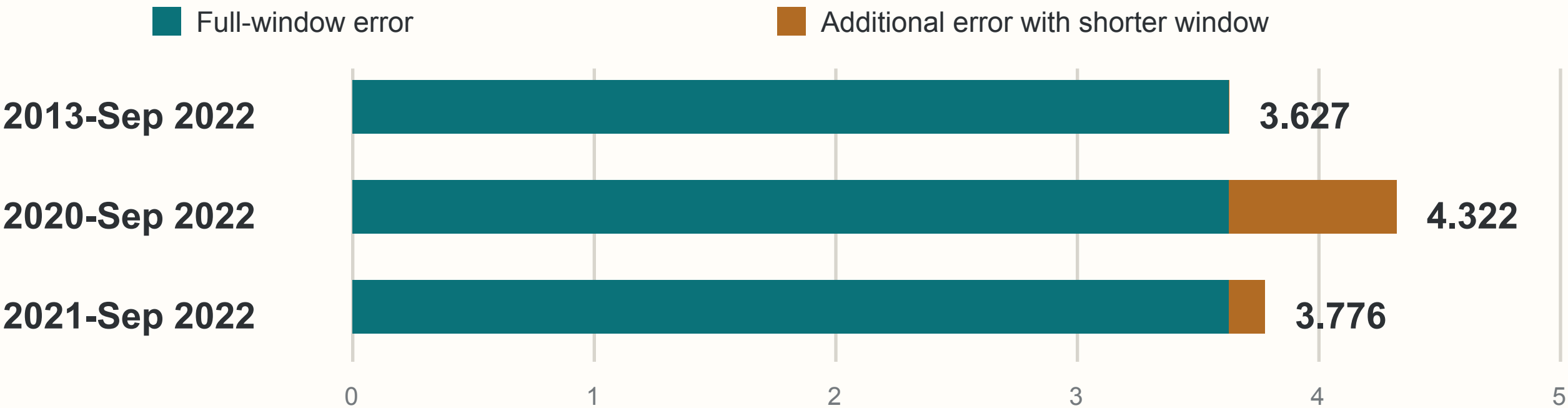
MAE: zero growth | RMSE: neural network | Review list: Ridge

13,490 complete histories with observed outcomes; lists recalculated at 10% within each quarter. Reused 2024 data, not a fresh test.  
Stochastic models: seed 42 shown. No other tested seed or version without deposit size exceeded Ridge. Anomalies do not establish distress.

**BUT WAIT... IN THESE VOLATILE ECONOMIC TIMES,  
WHAT IF WE TRAIN ON ONLY THE RECENT YEARS?**

Maybe older examples are less useful now. So I retrained starting in 2020, then 2021.  
Same model settings. Same 2023 validation. Same 13,532 bank-quarters scored in 2024.

BACK TO FORECAST ERROR: NEURAL NETWORK MAE (pp) · LOWER IS BETTER



**AND... THE ERRORS GOT BIGGER.**

Both shorter windows had higher MAE with all three seeds. Ridge had higher MAE too.

**So what if we keep the history, but let the forecast know the quarter?**

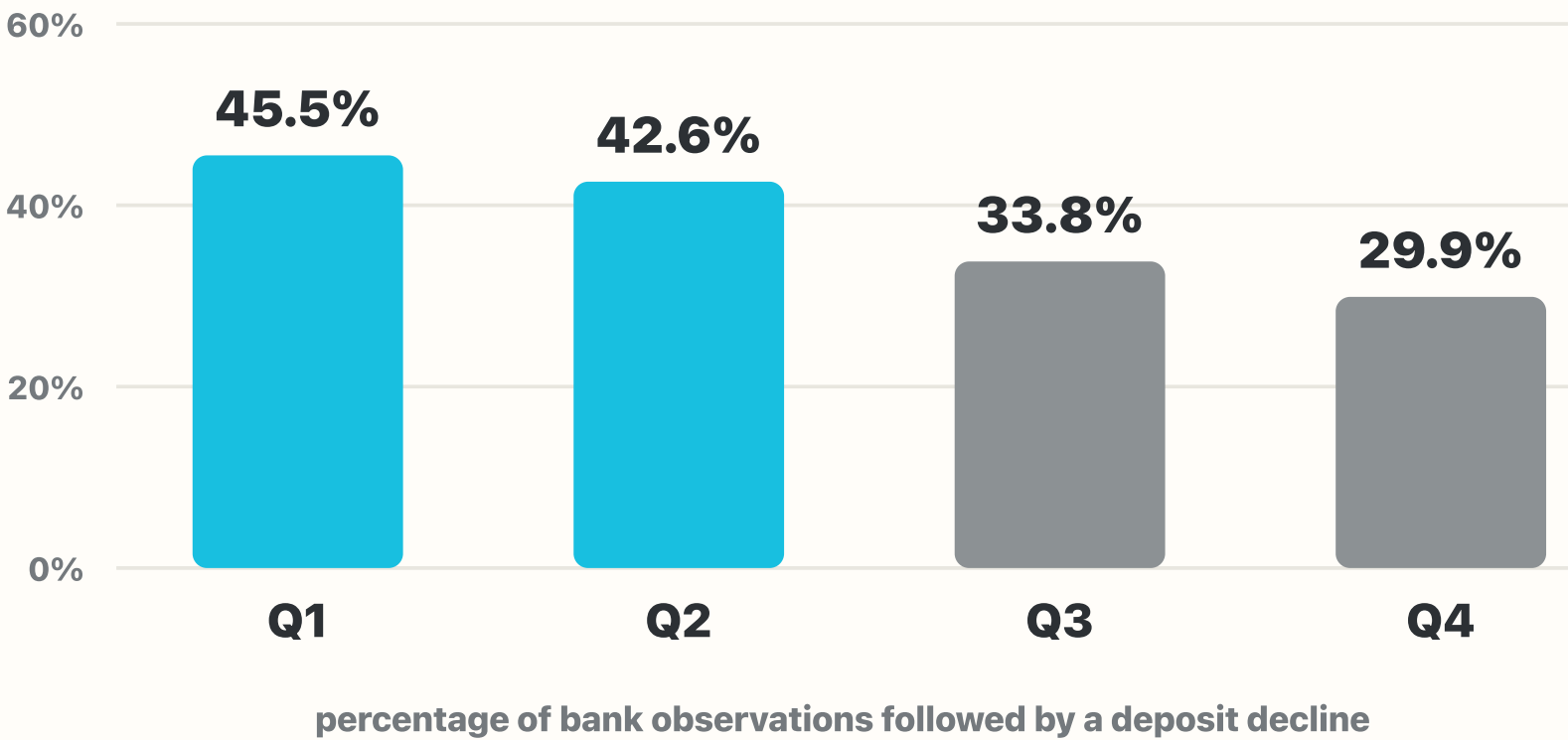
Seed 42 shown; also checked 7 and 99. Reused 2024 outcomes. Shorter windows also have fewer training examples; no regime effect is isolated.

# DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

**Q1: 45.5%** of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”

HOW OFTEN DID A BANK’S DEPOSITS FALL NEXT QUARTER? · TRAINING YEARS 2013–2022

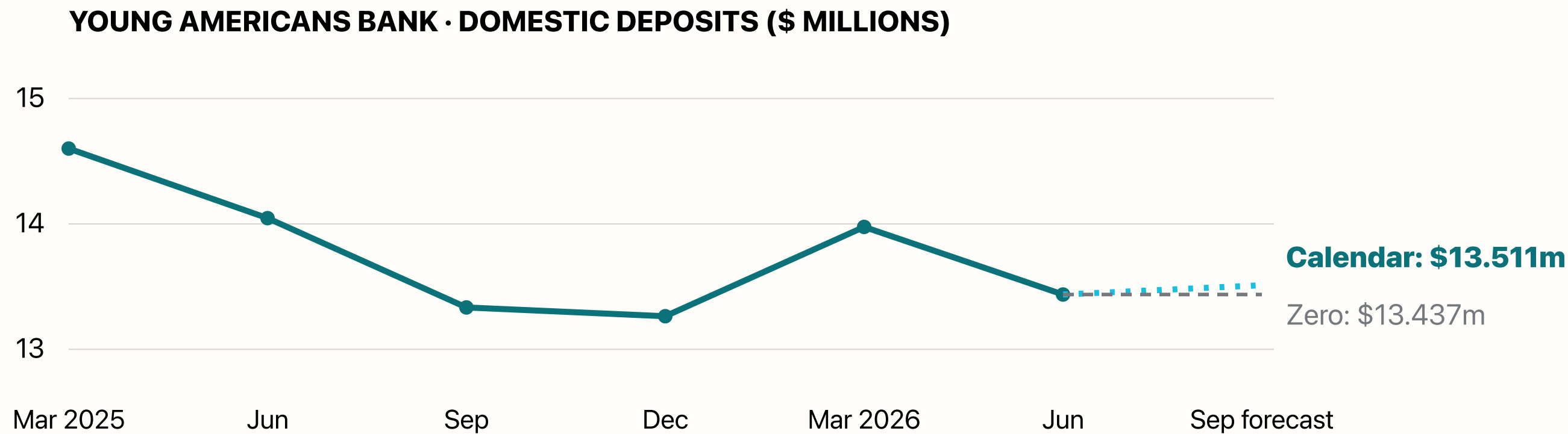


THE STARTING QUARTER MAY HELP SET A GROWTH FORECAST.

For each bar, count bank observations with a next-quarter decline, then divide by all observations starting in that quarter (training data through Sep 2022). Deposit dollars are not added together.

# MODEL CHOSEN FOR THE NEXT TEST: THE CALENDAR RULE. NOW WE WAIT FOR Q3 REPORTS.

Q3 ends September 30, 2026. Reports arrive afterward. The saved forecast is +0.550% per bank; zero is the benchmark.



Solid = reported balances. Dotted = calendar forecast (+0.550%). Dashed = no-change forecast.

**WHEN SEPTEMBER REPORTS ARRIVE, COMPARE MEAN ABSOLUTE ERROR.  
SAME BANKS. SAME QUARTER. LOWER MAE WINS.**

FDIC BankFind, CERT 27010, retrieved Sep 22, 2026. One-bank illustration of the saved Q2 median rule (+0.550399842%); test MAE across matched banks. Rule developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.